Save My Exams

GCSE · OCR · Computer Science

2 hours 33 questions

Additional Programming Techniques

Additional Programming Techniques

String Manipulation / File Handling / Records to Store Data / SQL / Arrays / Sub Programs / Random Number Generation

Easy (7 questions)	/9
Medium (14 questions)	/47
Hard (12 questions)	/55
Total Marks	**/111**

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Easy Questions

1 The function checkblock() checks whether a square on a board has been filled. When checkblock(4,2) is called, the value "A" is returned.

```
function checkblock(r,c)

    if gamegrid[r,c] == "A" or gamegrid[r,c] == "B" then

        outcome = gamegrid[r,c]

    else

        outcome = "FREE"

    endif

    return outcome

endfunction
```

State one feature of checkblock() that shows that it is a **function** and **not a procedure.**

..... **(1 mark)**

2 The algorithm for one section of a vending machine program is shown in pseudocode.

```
if money >= price then

  venditem()

  giveChange(money – price)

else

  print("Error – not enough money inserted")

endif
```

Give the identifier of **one variable** used in the algorithm.

..... **(1 mark)**

3 **What is the purpose** of upper() in string manipulation?

..... **(1 mark)**

4
```
01 message = "I LOVE COMPUTER SCIENCE"
02
```

Complete line 02 to output the message in **all lowercase**.

..... **(1 mark)**

5 **State** the primary function of the open() method in **file handling**?

..... **(1 mark)**

6 (a) **Write** a line of code to **create a 1D array** containing **10** elements.

The array should be named scores

..... **(1 mark)**

(b) **Write a line of code** to **add** the value **99** as the **third element** in scores 1D array

..... **(1 mark)**

7 **Define** the term 'record' in the context of a database.

.....

..... **(2 marks)**

Medium Questions

1 Customers at a hotel can stay between 1 and 5 (inclusive) nights and can choose between a basic room or a premium room.

A typical booking record is shown in the table:

firstName	Amaya
surname	Taylor-Ling
nights	3
room	Premium
stayComplete	False

Booking records are stored in a database table called TblBookings

The following SQL statement is written to display all customer bookings that stay more than one night.

SELECT ALL

FROM TblBookings

IF Nights < 1

The SQL statement is **incorrect**.

Rewrite the **SQL statement** so that it is correct.

.....

.....

.....

..... **(4 marks)**

2 The database table Results stores the results for each student in each of their chosen subjects.

StudentName	Subject	Grade
Alistair	English	3
Jaxon	Art	5
Alex	Art	4
Anna	French	7
Ismaael	Art	9

Complete the **SQL query** to return **all of the fields** for the **students** who take **Art**.

SELECT

FROM

WHERE

.....

.....

..... **(3 marks)**

3 Data for one week (Monday to Friday) is stored in a 2D array with the identifier minsPlayed.

The following table shows part of this array, containing 4 students.

			Students			
			Stuart	Wes	Victoria	Dan
			0	1	2	3
Days of the week	Mon	0	60	30	45	0
	Tue	1	180	60	0	60
	Wed	2	200	30	0	20
	Thu	3	60	10	15	15
	Fri	4	100	35	30	45

The teacher wants to output the number of minutes Dan (column index 3) played computer games on Wednesday (row index 2). The following code is written:

```
print(minsPlayed[3,2])
```

Write a line of code to **output the number of minutes** that **Stuart played computer games on Friday**.

You must use **either**:

- OCR Exam Reference Language, **or**
- a high-level programming language that you have studied.

..... **(1 mark)**

4 A program is being created to convert the data capacity of a storage device into a different measure.

The **function**, calculate(), takes the **measurement** (e.g. gigabytes) and the **number** (e.g. 2) as **two parameters**.

It then **returns the value in bits**.

The function returns **–1** if an invalid measurement was entered.

Complete the function calculate

```
function calculate(.......................,number)

if measurement == "gigabytes" then

   value = number * 1024 * 1024 * 1024 * 8

elseif measurement == "......................." then

   value = number * 1024 * 1024 * 8

elseif measurement = = "......................." then

   value = number * 1024 * 8

elseif measure = = "bytes" then

   value = number..................

else

   ...............................................

endif

return ...........................

endfunction
```

(6 marks)

5 OCRBlocks is a game played on a 5 × 5 grid. Players take it in turns to place blocks on the board. The board is stored as a two-dimensional (2D) array with the identifier gamegrid

Fig. 6.1 shows that players A and B have placed three blocks each so far.

	0	1	2	3	4
0	A			B	
1					
2		B			
3	A		B		
4			A		

Fig. 6.1

The function checkblock() checks whether a square on the board has been filled. When checkblock(4,2) is called, the value "A" is returned.

```
function checkblock(r,c)

  if gamegrid[r,c] == "A" or gamegrid[r,c] == "B" then

    outcome = gamegrid[r,c]

  else

    outcome = "FREE"

  endif

  return outcome

endfunction
```

Give the returned value when the following statements are called

Function call	Returned value
checkblock(2,1)	
checkblock(3,0)	
checkblock(2,3)	

.....

.....

..... **(3 marks)**

6 A vending machine stores the quantity of items available in a database table called ITEMS. The current contents of ITEMS is shown:

ItemCode	ItemName	Stock
A1	Crisps, bacon flavour	6
A2	Crisps, salted	2
B1	Chocolate bar	12
C1	Apple pieces	18
C2	Raisins	7

Complete the following **SQL statement** to display the **item code** for **all items** that have **fewer than 10 in stock**.

SELECT

FROM

....

.....

.....

.....

..... **(4 marks)**

7 (a) An example of the data held in a system is shown below in **Fig. 1**:

StudentName	Detail	Points	LetterSent
Kirstie	Homework forgotten	−2	FALSE
Byron	Good effort in class	1	TRUE
Grahame	100% in a test	2	FALSE
Marian	Bullying	−3	TRUE

Fig. 1

The data shown above in Fig. 1 is stored in a database table called Conduct.

Write an SQL statement to **select the** StudentName field for **all records** that have **negative** Points.

.....

.....

..... **(3 marks)**

(b) **State** the **wildcard** that can be used in SQL **to show all fields** from a table.

..... **(1 mark)**

8 (a) **Functions** and **procedures** are both examples of **sub programs**.

Describe one difference between a function and a procedure.

.....

..... **(2 marks)**

(b) **Describe two benefits** to a programmer of **using sub programs**.

.....

.....

.....

..... **(4 marks)**

9 (a) The variable message is assigned a value.

message = "abcd1234"

Show the output when each of the statement below executes.

The first output has been completed for you.

1. print(message.length) = 8
2. print(message.upper)
3. print(message.left(4)
4. print(int(message.right(4))*2)

.....

.....

..... **(3 marks)**

(b) **Write** an algorithm in pseudocode to:

- store "Hello" in the variable word1
- store "Everyone" in the variable word2
- concatenate word1 and word2 to store "HelloEeveryone" in the variable message

.....

.....

..... **(3 marks)**

10
```
areaCode = phoneNumber.left(3)
if areaCode == "020" then
   valid = true
else
   valid = false
endif
```
Describe how .left() is being used for data validation

.....

..... **(2 marks)**

11
```
myFile = open("names.txt")
myFile.writeLine(newName)
myFile.close()
```
Identify two standard operations required to update an existing text file

.....

..... **(2 marks)**

12
```
array scores[5]
scores[2] = 15
```
Describe how an index is used to modify an element in a 1D array

.....

..... **(2 marks)**

13
```
n = random(1,10)
```

State the purpose of the two parameters in random(1,10)

.....

..... **(2 marks)**

14 A hospital database stores patient details. A single entry includes: PatientID (Integer), Name (String), and IsAdmitted (Boolean).

Describe one reason why a **record** is more suitable than a **1D array** for storing this information.

.....

..... **(2 marks)**

Hard Questions

1 (a) A Basic room costs £60 each night. A Premium room costs £80 each night.

Create a function, newPrice(), that takes the **number of nights** and the **type of room** as **parameters**, **calculates** and **returns** the **price to pay**.

You **do not** have to validate these parameters.

You must use **either**:

- OCR Exam Reference Language, or
- a high-level programming language that you have studied.

.....

.....

.....

..... **(4 marks)**

(b) **Write program code**, that uses newPrice(), to **output the price** of staying in a **Premium room** for **5 nights**.

You must use either:

- OCR Exam Reference Language, or
- a high-level programming language that you have studied.

.....

.....

..... **(3 marks)**

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2 A program is created to **convert hours and minutes** into a **total number of minutes**.

The teacher wants to create a **sub program to perform the calculation**.

The program has been started but **is not complete**.

Complete the design for the program.

```
hours = input("Please enter number of hours played")

minutes = input("Please enter number of minutes played")

finalTotal = ..................................................................

print(finalTotal)



function ..............................................................................................

    ..............................................................................

    ..............................................................................

    ..............................................................................

    ..............................................................................

endfunction
```

.....

.....

.....

..... **(4 marks)**

3 (a) A programmer declares the following variables.

```
first = "Computer Science"

second = "is great"
```

State the **output** from the following line of program code.

```
print(first.length)
```

..... **(1 mark)**

(b) **State** the **output** from the following line of program code.

```
print(second.length DIV 3)
```

..... **(1 mark)**

(c) **State** the **output** from the following line of program code.

```
print(3 ^ 2)
```

..... **(1 mark)**

4 (a) A programmer declares the following variables.

first = "Computer Science"

second = "is great"

Strings can be concatenated (joined together) using the + operator.

For example, print("Maths " + second) will output Maths is great

Use string manipulation with the variables first and/or second to produce the following **output**.

great

..... **(1 mark)**

(b) **Use string manipulation** with the variables first and/or second to produce the following **output**.

Computer

..... **(1 mark)**

(c) **Use string manipulation** with the variables first and/or second to produce the following **output**.

Science is great

..... **(1 mark)**

5 The number of goals scored in each football match is held in an array called goals. An example of this array is shown.

goals = [0, 1, 3, 0, 4, 5, 2, 0, 2, 1]

Elliott wants to **count how many** matches **end** with **0 goals**.

Complete the following pseudocode for an algorithm to count up how many matches with 0 goals are stored in the array and then print out this value.

```
nogoalscount = 0
for count = 0 to (goals.length-1)
    if goals[.............................] == 0 then
        nogoalscount ..................................................
    endif
next count
print(.................................................)
```

(3 marks)

6 (a) OCR Land is a theme park aimed at children and adults. Entrance tickets are sold online. An adult ticket to OCR Land costs £19.99, with a child ticket costing £8.99. A booking fee of £2.50 is added to all orders.

A function, ticketprice(), takes the number of adult tickets and the number of child tickets as parameters. It calculates and returns the total price to be paid.

Use **pseudocode** to **create an algorithm for the function** ticketprice().

.....

.....

.....

.....

.....

.....**(6 marks)**

(b) **Tick (✓) one box** to identify the **data type** of the value returned from the function ticketprice(), **justifying** your choice.

Data type of returned value	Tick (✓) one box
Integer	
Real	
Boolean	
String	

Justification	

.....

.....**(2 marks)**

7 (a) A library gives each book a code made from the first three letters of the book title in upper case, followed by the last two digits of the year the book was published.

For example, "Poetry from the War", published in 2012 would be given the code POE12.

Complete the following **pseudocode for a function** definition that will take in the **book title** and **year** as **parameters** and **return the book code**.

```
01 function librarycode(title, ..................................................)

02     parta = title.subString(0, ..................................................)

03     partb = year.subString(2, 2)

04     ................................................ parta.upper + partb

05 endfunction
```

(3 marks)

(b) **Use pseudocode** to write an algorithm that does the following :

- Inputs the title and year of a book from the user.
- Uses the librarycode function above to work out the book code
- Permanently stores the new book code to the text file bookcodes.txt

.....

.....

.....

.....

.....

..... **(6 marks)**

8 An algorithm stores the position of a character on a straight line as an integer. A user can move

the character left or right.

The following algorithm:

- generates one random number between 1 and 512 (inclusive) to store as the position
- prompts the user to input a direction to move (left or right)
- takes a direction as input until a valid direction is input.

```
p = random(1, 512)
print("The position is ", p)
a = ""
while a != "left" and a != "right"
  a = input("Enter direction, left or right")
endwhile
```

If the character moves left, 5 is subtracted from the position.

If the character moves right, 5 is added to the position

The position of the character can only be between 1 and 512 inclusive.

- **The function** moveCharacter()**:**
    - takes the direction (left or right) and current position as parameters
    - changes position based on direction
    - sets position to 1 if the new position is less than 1
    - sets position to 512 if the new position is greater than 512
    - returns the new position.

Complete the function moveCharacter()

```
function moveCharacter(direction, position)
```

.....

.....

.....

.....

.....

..... **(6 marks)**

9 Students take part in a sports day. The students are put into teams.

Students gain points depending on their result and the year group they are in. The points are added to the team score.

The team with the most points at the end of the sports day wins

The individual results for each student in each event are stored in a database.

The database table TblResult stores the times of students in the 100 m race. Some of the data is shown:

StudentID	YearGroup	TeamName	Time
11GC1	11	Valiants	20.3
10VE1	10	Super-Team	19.7
10SM1	10	Super-Team	19.2
11JP2	11	Champions	19.65

Complete the SQL statement to show the Student ID and team name of all students who are in year group 11

SELECT StudentID, ...

FROM .....

(4 marks)

10
```
myFile = open("log.txt")
myFile.writeLine(message)
// programmer forgets to close the file
```
Explain why failing to close a text file after writing is poor programming practice

.....

.....

..... **(3 marks)**

11
```
array board[8,8]
board[0,0] = "Rook"
```
State two benefits of using a 2D array to represent a chess board instead of 64 separate variables

.....

..... **(2 marks)**

12
```
drop = random(1,100)
if drop <= 5 then
   item = "Rare"
else
   item = "Common"
endif
```
Evaluate why random number generation is important for game security/fairness

.....

.....

..... **(3 marks)**